ProHub CCS Chassis Dynamometers

Mains Power Requirements – USA & Japan

1. Supply Location

Each dynamometer cabinet is supplied with a 6 m pre-installed power harness for connection to the building's mains supply. The supply point should be located on the driver's side of the modules:

- **Left-hand drive (USA):** supply on the left side of the modules.
- **Right-hand drive (Japan):** supply on the right side of the modules.

Position the isolator between the modules and the cabinet assembly, approximately in line with the driver's door when a vehicle is on the dynamometer.

2. Supply Options at a Glance

Configuration	Voltage	Frequency	Supply rating	Connections	Harness label
Three phase	200–230 V	50 / 60 Hz	80 A per phase	L1, L2, L3, Earth (no Neutral)	USA → 3P
Single phase	220 V	60 Hz	100 A	L1, L2, Earth	USA 1P ←

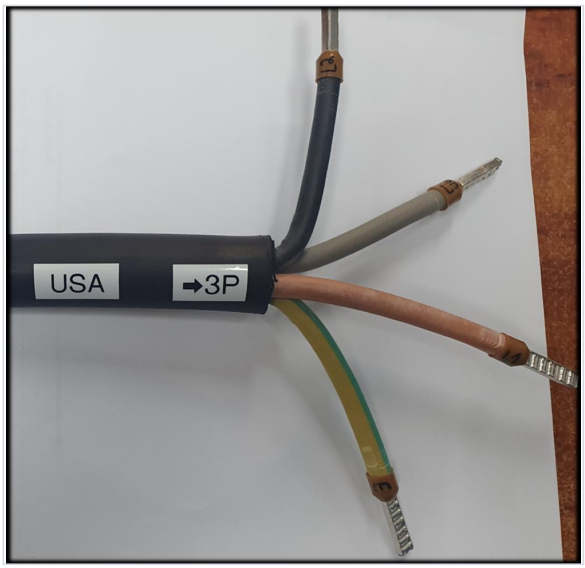
Isolation	Hardwire the harness to a wall-mounted on/off isolator protected by a C-curve circuit breaker rated for the supply (80 A per phase for three phase, 100 A for single phase).
Harness cable	4-core 10 mm² flexible cable, pre-installed and terminated inside the dynamometer cabinet.
Conductor colours	Earth – Green/Yellow · L1 – Brown · L2 – Black · L3 – Grey

IMPORTANT All mains connections must be made by a licensed electrician in accordance with local electrical codes and regulations.

3. Three Phase Supply – USA / Japan

Voltage	200–230 V, 3 phase
Frequency	50 / 60 Hz
Supply rating	80 A per phase
Configuration	4 wire: L1, L2, L3, Earth/Ground
Neutral	**Not used**

Connect the four conductors of the harness directly to the corresponding terminals of the wall isolator. The harness is labelled **USA → 3P** at the breakout.

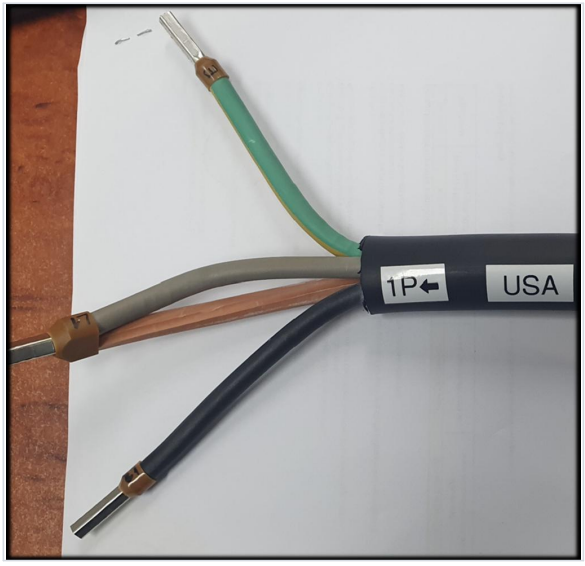


Three phase harness breakout (USA → 3P)

4. Single Phase Supply – USA

Voltage	220 V, single phase
Frequency	60 Hz
Supply rating	100 A
Configuration	3 wire: L1, L2, Earth/Ground

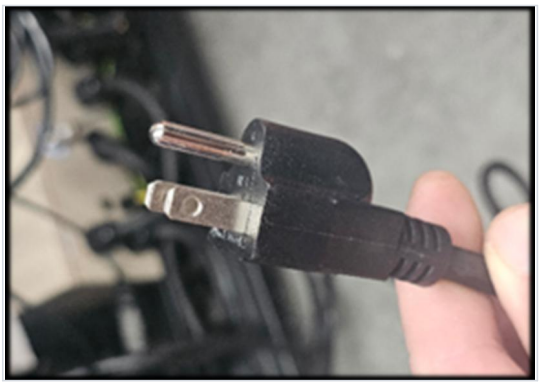
The single phase harness uses the same 4-core cable. The **Brown (L1)** and **Grey (L3)** conductors are joined in a common double ferrule and connected as L1. This allows quick conversion to three phase if required, simply by separating these two conductors. The harness is labelled **USA 1P ←**.



Single phase harness breakout (USA 1P ←)

5. 110 V Auxiliary Supply – US Models

US models include a 110 V power cable inside the cabinet. Route this cable down and out through the aperture at the bottom of the cabinet **before** fitting the CAN harnesses.



110 V auxiliary power cable

6. Optional Vehicle Cooling Fan Outlet

Where the optional vehicle cooling fan is purchased, a dedicated power outlet is fitted to the side of the dynamometer cabinet. The fan can be switched from the wireless keyboard or from the dynamometer software.

Fan option	Outlet
Three phase fan	4-pin outlet
Single phase fan	3-pin outlet



Single phase fan outlet (3-pin) on cabinet side

7. Wiring Diagrams

Detailed wiring diagrams for both supply configurations are provided on the following pages:

- **Diagram A** – USA / Japan three phase power specification (4 wires)
- **Diagram B** – USA / Japan single phase power specification (3 wires)

Diagram A – Three Phase Power Specification (4 Wires)

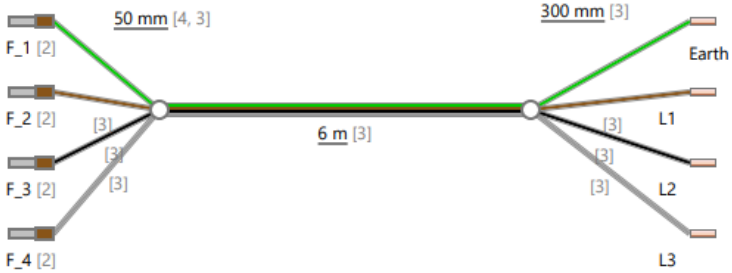
Revisions			
Rev.	Date	Author	Description
A	26/10/2020	Todd Lewis	

Notes

1. Hardwire to On/Off Isolator with min 80amp Breaker, C curve.



Wall Isolator Example



Connect this end to wall isolator

This end already terminated in cabinet.

This Cable is attached to the Dyno Cabinet, it is a 4 Core 10mm^2 flexible cable..

From	To	Conductor	Color	Gauge
F_1	Earth	CBL1.Green-Yellow		10 mm²
F_2	L1	CBL1.Brown-L1		10 mm²
F_3	L2	CBL1.Black-L2		10 mm²
F_4	L3	CBL1.Gray-L3		10 mm²

From	To	Conductor	Color	Gauge
Earth	F_1	CBL1.Green-Yellow		10 mm²
L1	F_2	CBL1.Brown-L1		10 mm²
L2	F_3	CBL1.Black-L2		10 mm²
L3	F_4	CBL1.Gray-L3		10 mm²

	Name	Date
Drawn By	----	----
Eng Approved	----	----
Mfg Approved	----	----
QA	----	----

MAINLINE CAN CONTROL SYSTEM

Title: USA/Japan 110v per phase, 200/208/220 3 Phase Dyno Cable
Part Number: ----
Description: AWD Hub Dyno Power Requiremnts and Wiring
Revision: ----

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Diagram B – Single Phase Power Specification (3 Wires)

